Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Application problems on Linear differential equations - Simple harmonic oscillation

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Application problems on Linear differential equations - Simple harmonic oscillation

1 Application problems on Linear differential equations - Simple harmonic oscillation

Study focus: Application problems on Linear differential equations - Simple harmonic oscillation. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000250 · EV0000251

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Linear differential equations have a wide variety of applications in science and engineering.

Key details

- They are used to model different situations, different phenomena in physics and engineering.
- A object with a mass attached to a vertical string which is oscillating to and fro.
- How these two situations can be modeled by second order linear differential equations.
- We know methods to solve second order linear differential equation.
- We solve the differential equation and we will learn more about the two different phenomena.
- And the restoring force of the spring is pushing it upwards.
- When the two forces are equal, the spring comes to an equilibrium position.
- Or if it is pushed upwards, it starts oscillating up and down.
- In the first diagram, there is nothing happening, a spring is just attached to a support.
- In the second diagram, an object with mass m is attached to a spring which is hung from a level surface.
- So because of the gravity, the mass is pulled downwards, but the restoring force is pushing it upwards and it comes from the same.
- To equilibrium position.
- This is force due to gravity.
- This is because of the mass M.
- And K into S.
- K is the force constant.

- We know it is spring constant.
- K into S is the displacement of the spring from the natural position.
- Because of the mass m mg is equal to case.
- It represents the displacement of the mass from the equilibrium position.
- regulatory motion can be represented by a second order linear differential equation.
- One is the force due to gravity, and the second one is the restoring force.
- We know Hook's law.
- What is from Hooke's Law?
- Hook's Law states that there.
- Restoring force of the spring is proportional to the displacement.
- A restoring force is equal to some constant into S plus X.
- It is a total displacement, and K is the proportionality constant, which is called as the force.
- Constant or the spring constant.
- And negative sign here indicates that the restoring force is in the opposite direction of the displacement.
- We know this.
- It is minus k into s plus x.
- The total the fourth I mean the restoring force acting on the system.
- system is minus k into s plus x.
- What is Newton seconds law?
- The force acting on the system is equal to m into a where a stands for the acceleration.
- M into a now.
- Equating the two, the total forces acting on the system is minus force due to gravity, and the second one is the restoring force.

Example or application from the lecture

- Consider a mechanical oscillator and an electrical oscillator.
- And consider an electrical circuit containing an inductor, capacitor, and resistor in series with a source of EMF.
- Let us see how.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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Core explanation · 05:00–09:59

A is the acceleration.

Key details

- This mg and k s gets cancelled and it becomes.
- M A is equal to minus k into x.
- But we know that the acceleration is a second order derivative of displacement with respect to time t.
- Putting this back in the equation and simplifying the equation becomes d square x by dt square.

- square plus k by m into x is equal to 0.
- It is a homogeneous differential equation, linear differential equation with constant coefficients.
- The auxiliary equation, which is associated with the differential.
- equation find out the roots and write the complementary function.
- It is actually square root of omega square minus omega square which is plus r minus i omega.
- The two roots are complex.
- It is C1 cos omega T.
- plus C2 sin omega T.
- X of T is equal to C1 cos omega T plus C2 sin omega T.
- What is X of T?
- It is the displacement of the mass from its equilibrium position and this equents a simple harmonic oscillation, simple harmonic motion.
- Introducing another variable phi.
- Put this back and simplify, we get x of t is equal to a into cos of omega t plus phi.
- Where a is equal to square root of C1 square by C2 square and tan phi is equal to minus C2 by C1.
- A represent A which is equal to square root of C1 square by C2 square represent the amplitude of the harm.
- motion and it gives the maximum displacement of the mass from its equilibrium position.
- It is a time taken for one complete oscillation.
- And we'll find out the natural frequency of the harmonic oscillator, which is a reciprocal.
- Of the period.
- It is square root of k by m.
- Equation which we derive for this simple harmonic motion.
- A spring is such that it would be stretched to 6 inches by 12 pound weight.
- Find also the amplitude and the period.
- The weight of the object is 12 pound.
- So, because of the weight, it is stretched to 6 inches.
- S is equal to 6 inches, and W which is equal to Mg is equal to 12 pound.
- We know the differential equation.
- It is d square x by dt square plus k by m x is equal to 0.
- Now first we need to find out this k the force constant k and the.

Example or application from the lecture

- The period of this oscillatory motion is given by 2 pi into square root of m by k.
- Consider see a big question.
- If the weight is pulled down 4 inches below the equilibrium point and given an upward velocity of 2 feet per second, determine the motion of the weight assuming no damping, no forces, external forces acting on the system.

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Core explanation · 10:00–14:59

Pass of the object M.

Key details

- K and M.
- We have MG is equal to MG is equal to.
- Then this is the weight of the object 12 pounds.
- or 6 inches is equal to 0.5 feet or 1 by 2 feet.
- It is equal to Mg by S.
- which what is m we have derived it is 3 by 8 into this is acceleration due to gravity 32 divided by the displacement of the mass because of the weight which is equal to 6.
- which is equal to how much?
- We found out Kn.
- Omega is equal to 8.
- General solution C1 cos omega t plus C2 sin omega T.
- Did you understand how we got this?
- Because omega square is equal to 64, omega is equal to 8, and the general solution we wrote.
- It is C1 cos omega t plus C2 sin omega T.
- Now the next question is how do we find the constant C1 and C2?
- To find the constant C1 and C2, we need to use the data.
- To find C1 and C2, we will use the initial condition.
- Sine 0 is 0, cos 0 is 1, c1 is equal to 1 by 3.
- What is velocity? dx by dt.
- We get dx by dt is equal to 1 by 3 into cos differentiation is minus sine into 8 t differentiation is 8.
- But the negative sign, why is this negative sign?
- Opposite direction of the displacement.
- If it is downward velocity, we use the sign plus.
- We'll give positive negative sign.
- C2 is equal to minus 1 by 4.
- Say dx by dt is minus 2 and t is equal to 0.
- This term by and you get C2 into A t is equal to now minus 2.
- It is 1 by 3 cos C8 minus 1 by 4 sin A T.

Example or application from the lecture

- So the weight of the mass, weight of the mass we know it is equal to m into g, where g is the acceleration due to gravity because here it is given in feed.
- Let us write the gravitational mean acceleration due to gravity.
- M into g s is known, s is given from the question.

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Core explanation · 15:00–19:59

X of T, it is a function of T time.

Key details

- is 1 by 3 and C2 is minus 1 by 4, simplify amplitude is equal to 5 by 12.
- And also what is the next question?
- Period the motion which is 2 pi by omega.
- But what is omega?
- Omega is equal to we have derived omega is equal to 8.
- This is how we find the period of the motion amplitude by writing the equation for the displacement XFP.

Example or application from the lecture

- The amplitude is given by this is the formula square root of C1 square plus C2 square.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Application problems on Linear differential equations - Simple harmonic oscillation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Application problems on Linear differential equations - Simple harmonic oscillation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Application problems on Linear differential equations - Simple harmonic oscillation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9983; only the JSON segmentation supplies the displayed timestamps.
Application problems on Linear differential equations - Simple harmonic oscillation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00062	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0056. Sidecar variants compare...
SRC00063	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00064	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0056. Sidecar variants compare...
SRC00065	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0056. Sidecar variants compare...

C Revision and retrieval practice